1 Realization of Differential Forms

1.1 Cotangent Spaces

Definition 1.1 (Cotangent Spaces)**.**
The cotangent space $T_p^*\mathcal{M}$ at $p \in \mathcal{M}$ is the set of all linear functions $f : T_p\mathcal{M} \to \mathbb{R}$.
Its member is called a cotangent vector.
$\dim T_p^*\mathcal{M} = \dim T_p\mathcal{M}$.

Definition 1.2 (One-Form)**.**
A one-form on $\mathcal{M}$ is a smooth assignment of cotangent vectors $\omega : p \mapsto \omega_p$.
It may be understood as a covector field.

Definition 1.3 (Basis Cotangent Vectors)**.**
The basis cotangent vectors is chosen to be the dual basis of the basis tangent vectors **??**,

$$(dx^\mu)_p((\partial_\nu)_p) = \delta^\mu{}_\nu.$$

Theorem 1.1 (Coordinate Expression of Cotangent Vectors)**.**
Any $f \in T_p^*\mathcal{M}$ can be expanded as

$$f = f_\mu (dx^\mu)_p.$$

Any one-form ω can be expressed as

$$\omega = \omega_\mu dx^\mu.$$

Theorem 1.2 (Pullback as Dual of Pushforward)**.**
Given two manifolds $\mathcal{M}$ with dimension m and $\mathcal{N}$ with dimension n and a C^∞ function $f : \mathcal{M} \to \mathcal{N}$, the pullback of a one-form is the dual of pushforward. That is,

$$(h^*\omega)_p(v) := \omega_p(h_* v).$$

Theorem 1.3 (Local Representative of Pullback)**.**
Let $\dim \mathcal{M} = m, \dim \mathcal{N} = n$, $h : \mathcal{M} \to \mathcal{N}$, $\{x^1, \dots, x^m\}$ be the local coordinates of $\mathcal{M}$ around p, and $\{y^1, \dots, y^n\}$ be the local coordinates of $\mathcal{N}$ around $h(p)$. Then

$$h^*\omega = \sum_{\mu=1}^{m} \sum_{\nu=1}^{n} \omega_\nu \left.\frac{\partial h^\nu}{\partial x^\mu}\right|_p (dx^\mu)_p,$$

where $J^\nu{}_\mu := \left.\frac{\partial h^\nu}{\partial x^\mu}\right|_p := \left(\partial_\mu\right)_p (y^\nu \circ h)$ is the Jacobian matrix.

Definition 1.4 (n-Forms)**.**
An n-form is a tensor field of type $(0, n)$ that is totally skew-symmetric (or alternating, or totally antisymmetric), i.e.,

$$\omega(X_1, X_2, \dots, X_n) = (\operatorname{sgn} \sigma)\omega(X_{\sigma(1)}, \dots, X_{\sigma(n)}),\ \forall \sigma \in S_n.$$

The set of all n-forms on $\mathcal{M}$ is denoted as $\Lambda^n(\mathcal{M})$. The set of all forms is $\Lambda(\mathcal{M}) = \bigoplus_{n=0}^{\dim \mathcal{M}} \Lambda^n(\mathcal{M})$. Conventionally, we classify functions as 0-forms.

1.2 The Exterior Product

1.2.1 Definition and Properties

Definition 1.5 (Exterior Product)**.**
Given $\omega_1 \in \Lambda^{n_1}(\mathcal{M}), \omega_2 \in \Lambda^{n_2}(\mathcal{M})$, their exterior product is a $(n_1 + n_2)$-form given by,

$$\omega_1 \wedge \omega_2 := \frac{1}{n_1!n_2!} \sum_{\sigma \in S_{n_1+n_2}} (\operatorname{sgn} \sigma)(\omega_1 \otimes \omega_2)_\sigma.$$

Written explicitly,

$$(\omega_1 \wedge \omega_2)(X_1, \ldots, X_{n_1+n_2}) := \frac{1}{n_1!n_2!} \sum_{\sigma \in S_{n_1+n_2}} (\operatorname{sgn} \sigma)(\omega_1 \otimes \omega_2)(X_{\sigma(1)}, \ldots, X_{\sigma(n_1+n_2)})$$

Remark.
I'll take the alternating property and associativity of the exterior product for granted. For a detailed proof, see Hoffman. ♍

Theorem 1.4 (Commutativity with Pullback)**.**
Given $h : \mathcal{M} \to \mathcal{N}$ and $\alpha, \beta \in \Lambda(\mathcal{N})$, then

$$h^*(\alpha \wedge \beta) = (h^*\alpha) \wedge (h^*\beta).$$

Remark.
For a "generalized" pullback, we have,

$$(h^*(\alpha))(X_1, \dots, X_{n_1}) = \alpha(h_* X_1, \dots, h_* X_{n_1}).$$

♍

Proof.

$$\begin{aligned}
&(h^*\alpha)\wedge(h^*\beta) \\
&\quad = \frac{1}{n_1! n_2!} \sum_{\sigma \in S_{n_1+n_2}} (\operatorname{sgn}\sigma) \alpha \otimes \beta(h_* X_{\sigma(1)}, \dots, h_* X_{\sigma(n_1+n_2)}). \\
&\quad = \frac{1}{n_1! n_2!} \sum_{\sigma \in S_{n_1+n_2}} (\operatorname{sgn}\sigma) h^* \left(\alpha \otimes \beta(X_{\sigma(1)}, \dots, X_{\sigma(n_1+n_2)})\right). \\
&\quad = h^* \left(\frac{1}{n_1! n_2!} \sum_{\sigma \in S_{n_1+n_2}} (\operatorname{sgn}\sigma) \alpha \otimes \beta(X_{\sigma(1)}, \dots, X_{\sigma(n_1+n_2)}) \right). \\
&\quad = h^*(\alpha \wedge \beta).
\end{aligned}$$

▨

Theorem 1.5 (Skew-Symmetry)**.**
The exterior product makes $\Lambda(\mathcal{M})$ a graded algebra with skew-symmetry given by

$$\omega_1 \wedge \omega_2 = (-1)^{n_1 n_2} \omega_2 \wedge \omega_1.$$

Proof. In the definition of exterior product, first fix $\sigma = \sigma_0$ to consider only one term.

When we switch ω_1 and ω_2, we are essentially doing

$$\begin{aligned}
&(\omega_2 \otimes \omega_1)(X_{\sigma_0(1)}, \dots, X_{\sigma_0(n_2)}, \underbrace{X_{\sigma_0(n_2+1)}, \dots, X_{\sigma_0(n_1+n_2)}}) \\
=&(\omega_1 \otimes \omega_2)(\underbrace{X_{\sigma_0(n_2+1)}, \dots, X_{\sigma_0(n_1+n_2)}}, X_{\sigma_0(1)}, \dots, X_{\sigma_0(n_2)}).
\end{aligned}$$

Now,

$$\underbrace{1, 2, \dots, n_2}, \underbrace{n_2 + 1}, \dots, n_1 + n_2$$

$$\downarrow \; n_2 \text{ times}$$

$$\underbrace{n_2 + 1}, \underbrace{1, 2, \dots, n_2}, \dots, n_1 + n_2$$

$$\downarrow \; (n_1 - 1)n_2 \text{ times}$$

$$\underbrace{n_2 + 1, \dots, n_1 + n_2}, \underbrace{1, 2, \dots, n_2}$$

So $n_1 n_2$ transposes can achieve the desired effect. Therefore, every term in the summation is multiplied by $(-1)^{n_1 n_2}$, and we get the desired result. ▨

Theorem 1.6 (Dimension of n-Forms)**.**
Let $\dim \mathcal{M} = m$. If $1 \leq n \leq m$, then $\Lambda^n(\mathcal{M}) = \binom{m}{n}$. If $n > m$, then $\Lambda^n(\mathcal{M}) = 0$.
Moreover, a basis for $\Lambda^n(\mathcal{M})_p$ is given by,

$$(dx^{\mu_1})_p \wedge (dx^{\mu_2})_p \wedge \cdots \wedge (dx^{\mu_n})_p, \; 1 \leq \mu_1 \leq \cdots \leq \mu_n \leq m.$$

Remark.
The proof is quite a pleasure to read (and to think of). Please see Hoffman. ♍

1.2.2 Evaluation

Theorem 1.7 (Evaluate k-forms with Vector Fields)**.**
On a smooth manifold $\mathcal{M}$, under the chart $\phi = (x^1, \dots, x^m)$, let k vector fields have local representation $X_b = \xi^a{}_b \partial_a$. Then

$$(dx^1 \wedge \cdots \wedge dx^k)(X_1, \dots, X_k) = \det \xi^a{}_b.$$

Proof.

$$
\begin{aligned}
&(dx^1 \wedge \cdots \wedge dx^k)(X_1, \dots, X_k) \\
=&(dx^1 \wedge \cdots \wedge dx^k)({\xi^{i_1}}_1 \partial_{i_1}, \dots, {\xi^{i_k}}_k \partial_{i_k}) \\
=&{\xi^{i_1}}_1 {\xi^{i_k}}_k (dx^1 \wedge \cdots \wedge dx^k)(\partial_{i_1}, \dots, \partial_{i_k}).
\end{aligned}
$$

To evaluate the last term, we first see that only terms with $i_\mu \neq i_\nu$ appear. Thus, we rewrite it using permutations,

$$
\begin{aligned}
&{\xi^{i_1}}_1 {\xi^{i_k}}_k (dx^1 \wedge \cdots \wedge dx^k)(\partial_{i_1}, \dots, \partial_{i_k}) \\
=&\sum_{\sigma \in S_k} {\xi^{\sigma(1)}}_1 {\xi^{\sigma(k)}}_k (dx^1 \wedge \cdots \wedge dx^k)(\partial_{\sigma(1)}, \dots, \partial_{\sigma(k)}).
\end{aligned}
$$

Moreover, since exterior product is alternating,

$$
\begin{aligned}
&\sum_{\sigma \in S_k} {\xi^{\sigma(1)}}_1 {\xi^{\sigma(k)}}_k (dx^1 \wedge \cdots \wedge dx^k)(\partial_{\sigma(1)}, \dots, \partial_{\sigma(k)}) \\
=&\sum_{\sigma \in S_k} (\operatorname{sgn} \sigma) {\xi^{\sigma(1)}}_1 \dots {\xi^{\sigma(k)}}_k \\
=&\det {\xi^a}_b.
\end{aligned}
$$

∎

1.3 The Exterior Derivative

Definition 1.6 (Exterior Derivative)**.**
Let a k-form be $\omega = \omega_{i_1 \dots i_k} dx^{i_1} \wedge \dots \wedge dx^{i_k}$. Then

$$d\omega := \left(\frac{\partial \omega_{i_1 \dots i_k}}{\partial x^{i_0}} dx^{i_0} \right) \wedge dx^{i_1} \wedge \dots \wedge dx^{i_k}.$$

If $\omega \in \Lambda^{\dim \mathcal{M}}(\mathcal{M})$, we define $d\omega = 0$.

Theorem 1.8.
A useful formula for the evaluation of the exterior derivative of a 0-form $f \in C^\infty(\mathcal{M})$ is the following,

$$df(X) := \mathcal{L}_X f = Xf.$$

Proof. In coordinates,

$$df = (\partial_\mu f)(dx^\mu).$$

So

$$df(X) = (\partial_\mu f)(dx^\mu)(X) = X^\mu(\partial_\mu f) = (X^\mu \partial_\mu) f = Xf.$$

▨

Theorem 1.9 (Invariant Formula of Exterior Derivative for 1-forms)**.**
In particular for a 1-form ω,

$$d\omega(X, Y) = \mathcal{L}_X(\omega(Y)) - \mathcal{L}_Y(\omega(X)) - \omega([X, Y]).$$

Theorem 1.10 (Exterior Derivative and Product)**.**

$$d(\omega_1 \wedge \omega_2) = d\omega_1 \wedge \omega_2 + (-1)^{\deg \omega_1} \omega_1 \wedge d\omega_2.$$

Theorem 1.11 (Exterior Derivative and Pullback)**.**
Given $h : \mathcal{M} \to \mathcal{N}$, ω an n-form on $\mathcal{N}$, then

$$d(h^* \omega) = h^*(d\omega).$$

Theorem 1.12 (Functional Linearity of Exterior Derivative)**.**
Let ω be a 1-form on $\mathcal{M}$. Then $d\omega$ satisfies,

$$d\omega(fX, Y) = f d\omega(X, Y), \ \forall f \in C^\infty(\mathcal{M}),$$

where fX is a vector field that gives $(fX)(p) = f(p)X_p$.

Proof. By Theorem 1.9,

$$d\omega(fX, Y) = \mathcal{L}_{fX}(\omega(Y)) - \mathcal{L}_Y(\omega(fX)) - \omega([fX, Y]).$$

We break it down term by term. Firstly,

$$(\mathcal{L}_{fX}(\omega(Y)))(p) = f(p)X_p(\omega(Y)) = f(p)(\mathcal{L}_X(\omega(Y)))(p).$$

So

$$\mathcal{L}_{fX}(\omega(Y)) = f \cdot \mathcal{L}_X(\omega(Y)).$$

Secondly, we tackle $\mathcal{L}_Y(\omega(fX))$. In particular,

$$\omega(fX)(p) = \omega_p(f(p)X_p) = f(p)\omega_p(X_p) = f(p)(\omega(X))(p).$$

Therefore,

$$\mathcal{L}_Y(\omega(fX)) = \mathcal{L}_Y(f \cdot \omega(X)) = (\mathcal{L}_Y f)\omega(X) + f \cdot \mathcal{L}_Y(\omega(X)).$$

Thirdly,

$$\omega([fX, Y]) = \omega((fX) \circ Y - Y \circ (fX)).$$

In particular,

$$((Y \circ (fX))(g))(p) = Y_p((fX)(g)) = Y_p(f \cdot Xg) = (Y_p f)((Xg)(p)) + f(p) \cdot Y_p(Xg).$$

So,

$$Y \circ (fX) = (\mathcal{L}_Y f)X + f \cdot Y \circ X.$$

Substituting back,

$$\begin{aligned}
\omega([fX, Y]) &= \omega(f \cdot X \circ Y - (\mathcal{L}_Y f)X - f \cdot Y \circ X) \\
&= \omega(f[X, Y] - (\mathcal{L}_Y f)X) \\
&= f\omega([X, Y]) - (\mathcal{L}_Y f)\omega(X).
\end{aligned}$$

Finally,

$$\begin{aligned}
d\omega(fX,Y) &= \mathcal{L}_{fX}(\omega(Y)) - \mathcal{L}_Y(\omega(fX)) - \omega([fX,Y]) \\
&= f\cdot\mathcal{L}_X(\omega(Y)) - (\mathcal{L}_Y f)\omega(X) - f\cdot\mathcal{L}_Y(\omega(X)) - f\omega([X,Y]) + (\mathcal{L}_Y f)\omega(X) \\
&= f(\mathcal{L}_X(\omega(Y)) - \mathcal{L}_Y(\omega(X)) - \omega([X,Y])) \\
&= f d\omega(X,Y).
\end{aligned}$$

▨

Corollary 1.12.1 (Local Nature of Exterior Derivative)**.**
When ω is fixed, the value of $d\omega$ depends only on the local values of vector fields.

$$d\omega(X,Y)(p) = X^\mu(p)Y^\nu(p)d\omega(\partial_\mu,\partial_\nu)(p).$$

Proof. Write $X = X^\mu\partial_\mu$, noting that $X^\mu \in C^\infty(\mathcal{M})$, and use Theorem 1.12. ▨

1.4 DeRham Cohomology

Theorem 1.13 (Twice Exterior Differential)**.**
For all $\omega \in \Lambda^n(\mathcal{M}), 1 \leq n \leq \dim M$, we have

$$d^2\omega = 0.$$

Remark.
This means

$$\mathrm{Im}(d : \Lambda^{n-1}(\mathcal{M}) \to \Lambda^n(\mathcal{M})) \subseteq \mathrm{Ker}(d : \Lambda^n(\mathcal{M}) \to \Lambda^{n+1}(\mathcal{M})).$$

This type of structure is called a differential complex, and is common in many structures. ♍

Definition 1.7 (Closed Form)**.**
An n-form ω is closed if $d\omega = 0$. The set of all closed n-forms is denoted $Z^n(\mathcal{M})$.

Definition 1.8 (Exact Form)**.**
An n-form ω is exact if $\omega = d\beta$ for some $(n-1)$-form β. The set of all exact n-forms is denoted $B^n(\mathcal{M})$.

Remark.
It is guaranteed that $B^n(\mathcal{M}) \subseteq Z^n(\mathcal{M})$, that is, exactness implies closure. But how much closed form is not exact is the study of cohomology theory. ♍

Theorem 1.14 (Poincare's Lemma)**.**
On Euclidean space $\mathbb{R}^m$,

$$B^n(\mathcal{M}) = Z^n(\mathcal{M}), \ \forall n > 0.$$

Definition 1.9 (DeRham Cohomology Groups)**.**
The DeRham cohomology groups $H^n(\mathcal{M}), 0 \leq n \leq \dim \mathcal{M}$ are the quotient

spaces

$$H^n(\mathcal{M}) := Z^n(\mathcal{M})/B^n(\mathcal{M}).$$

Remark.

Recall the definition of quotient groups that $H^n(\mathcal{M})$ consists of elements of form $z + B^n(\mathcal{M}), z \in Z^n(\mathcal{M})$.

If all closed forms are exact, $Z^n(\mathcal{M}) \subseteq B^n(\mathcal{M})$, then $H^n(\mathcal{M}) \simeq \{0\}$. ♍

Theorem 1.15 (Criterion of Exact ODE)**.**

On the Euclidean space $\mathbb{R}^2$, given a 1-form $\omega = \omega_1 dx^1 + \omega_2 dx^2$. Then

$$\omega \in B^1(\mathbb{R}^2) \iff \partial_2 \omega_1 = \partial_1 \omega_2.$$

Remark.

This is an important theorem to me, for it connects the "exactness of differential forms" to the familiar notion of "exactness of differential equations".

It also provides the first hints that we are actually integrating forms, and that exterior differentiation of a 0-form resembles gradient in usual vector calculus terms. ♍

Proof. Via Poincare lemma Theorem 1.14, on $\mathbb{R}^2$, exactness is equivalent to closure. So we need only to determine the condition that $d\omega = 0$. Using Definition 1.6,

$$\begin{aligned} d\omega &= \partial_\nu \omega_{\mu_1} dx^\nu \wedge dx^{\mu_1} \\ &= \partial_2 \omega_1 dx^2 \wedge dx^1 + \partial_1 \omega_2 dx^1 \wedge dx^2 \\ &= (\partial_2 \omega_1 - \partial_1 \omega_2)\, dx^2 \wedge dx^1. \end{aligned}$$

▨